FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Main)-2015

ANSWERS, HINTS & SOLUTIONS

FULL TEST – IV
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**B**	**D**	**B**
2.	**B**	**B**	**D**
3.	**A**	**A**	**D**
4.	**C**	**B**	**A**
5.	**B**	**D**	**A**
6.	**D**	**C**	**D**
7.	**C**	**A**	**B**
8.	**A**	**C**	**C**
9.	**D**	**B**	**A**
10.	**D**	**C**	**A**
11.	**D**	**B**	**D**
12.	**B**	**D**	**B**
13.	**C**	**C**	**D**
14.	**B**	**D**	**C**
15.	**A**	**C**	**D**
16.	**B**	**B**	**B**
17.	**B**	**D**	**B**
18.	**A**	**B**	**B**
19.	**D**	**C**	**A**
20.	**A**	**B**	**C**
21.	**D**	**C**	**D**
22.	**A**	A	**B**
23.	**C**	**D**	**A**
24.	**D**	**C**	**C**
25.	**D**	**B**	**A**
26.	**C**	**C**	**B**
27.	**A**	**C**	**C**
28.	**B**	**A**	**C**
29.	**D**	**D**	**A**
30.	**D**	**C**	**C**

Physics

PART – I

SECTION – A

1. $\vec{A}\times\vec{B}$ is a vector perpendicular to both $\vec{A}$ and $\vec{B}$

Now, $\vec{A}\times\vec{B}=\left(\hat{i}-2\hat{j}+\hat{k}\right)\times\left(3\hat{i}+\hat{j}-2\hat{k}\right)$

$3\hat{i}+5\hat{j}+7\hat{k}$

Now, $\hat{n}=\dfrac{\vec{A}\times\vec{B}}{\left|\vec{A}\times\vec{B}\right|}$

$=\dfrac{3\hat{i}+5\hat{j}+7\hat{k}}{\sqrt{3^2+5^2+7^2}}$

$=\dfrac{3\hat{i}+5\hat{j}+7\hat{k}}{\sqrt{83}}$

2. Horizontal velocity remains same

$30^o\cos 60^o=v\cos 45^o$

$\Rightarrow v = 15\sqrt{2}$ m/s

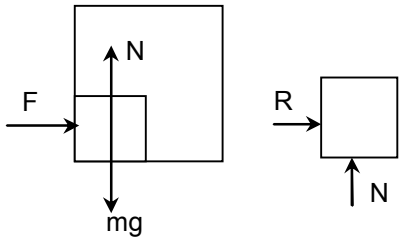
3. $v=\sqrt{29}\ t$

$a=\dfrac{dv}{dt}=\sqrt{29}$

$R=ma=\sqrt{29}$ N

$F_{net}=\sqrt{N^2+R^2}$

$=\sqrt{(10)^2+(29)}=\sqrt{129}$ N



4. Δp after 1st impact $=ep-(-p)$

$=p(1+e)$

Similarly Δp after 2nd impact = $ep(1+e)$

So, $p_{net}=p(1+e)\left[1+e+e^2+.....\right]$

$=\dfrac{p(1+e)}{(1-e)}$

5. Force of explosion is internal and system is initially at rest

Let the velocities of the first two fragments are $v\hat{i}$ & $v\hat{j}$ and that of the fragment 2m be $v_1\hat{i}+v_2\hat{j}+v_3\hat{k}$,

So, $\vec{p}_1+\vec{p}_2+\vec{p}_3=0$

$\Rightarrow mv\hat{i}+mv\hat{j}+2m\left(v_1\hat{i}+v_2\hat{j}+v_3\hat{k}\right)=0$

$\Rightarrow v_1=-\dfrac{v}{2},\ v_2=-\dfrac{v}{2}$ and $v_3=0$

$$\vec{p}_3 = -2m\left(\frac{v}{2}\hat{i} + \frac{v}{2}\hat{j}\right)$$

So, $K_f = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}(2m)\left(v_1^2 + v_2^2 + v_3^2\right)$

$= \frac{3}{2}mv^2$

Energy released in explosion

$\Delta E = K_f - K_i$

$= \frac{3}{2}mv^2 - 0 = \frac{3}{2}mv^2$

6. Angular momentum about point of contact will be conserved

$L_i = L_f$

$$\left(\frac{2}{5}Mr^2\right)\frac{v_0}{2r} + Mrv_0 = \left(\frac{2}{5}Mr^2\right)\frac{v}{r} + Mrv$$

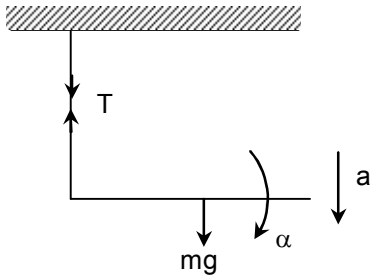
$\Rightarrow v = \frac{6}{7}v_0$

7. $mg - T = ma$ and

$$\frac{mgl}{2} = I\alpha = \frac{ml^2}{3} \times \frac{a}{\left(\frac{l}{2}\right)}$$

$\Rightarrow a = \frac{3g}{4}$

So, $T = \frac{mg}{4}$



8. $I = -\frac{K}{r} = -\frac{dV}{dr}$

$$\Rightarrow \int_{V_0}^{V} dV = K\int_{r_0}^{r}\frac{dr}{r}$$

$$\Rightarrow V = K\ln\frac{r}{r_0} + V_0$$

9. By conservation of angular momentum, $mv_1R = mv_2r$

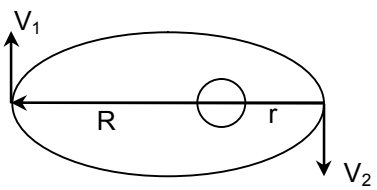
$\Rightarrow \quad v_1R = v_2r$

By conservation of energy,

$$-\frac{GMm}{R} + \frac{1}{2}mV_1^2 = -\frac{GMm}{r} + \frac{1}{2}mV_2^2$$

$$\Rightarrow V_1 = \sqrt{\frac{2GMr}{R(R+r)}}$$

Now, $mv_1R = m\sqrt{\frac{2GMrR}{r+R}}$



10. By equation of continuity

$$Av = av'$$

$$\Rightarrow v' = \left(\frac{A}{a}\right)v = \left(\frac{\pi R^2}{\pi r^2}\right)\times v$$

$$\Rightarrow v' = 400 \text{ cm/sec.}$$

11. For cylinder A ;

$$\Delta Q = nC_p\Delta T_1$$

For cylinder B, $\Delta Q = nC_v\Delta T_2$

Hence, $nC_p\Delta T_1 = nC_v\Delta T_2$

$$(C_v + R)30 = C_v\Delta T_2$$

For diatomic gas,

$$C_v = \frac{5}{2}R$$

$$\Rightarrow \Delta T_2 = 42 \text{ K}$$

12. $\frac{dT}{dt} \propto \Delta\theta$ $\qquad$ $\frac{d\theta}{dt} \propto (\theta - \theta_0)$

$$\Rightarrow dT = K\Delta\theta dt$$

In first case,

$dT = 61 - 59 = 2^oC$; $\Delta\theta = 30^oC$, dt = 4 min.

For second case, $dT = 2^oC$; $\Delta\theta = 20^oC$

$$K = \frac{dT}{\Delta\theta dt} = \frac{2}{30\times 4} = \frac{1}{60}$$

$$dt = \frac{dT}{K\Delta\theta} = \frac{2}{\frac{1}{60}\times 20} = 6 \text{ min.}$$

13. $y = a(\sin\omega t + \cos\omega t)$

$$= a\sqrt{2}\left(\sin\omega t\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\cos\omega t\right)$$

$$= a\sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)$$

The motion is SHM with amplitude $a\sqrt{2}$

14. $$\frac{aT}{x} = \frac{\omega^2 xT}{x} = \frac{4\pi^2}{T^2}\times T$$

$$= \frac{4\pi^2}{T} = \text{Constt.}$$

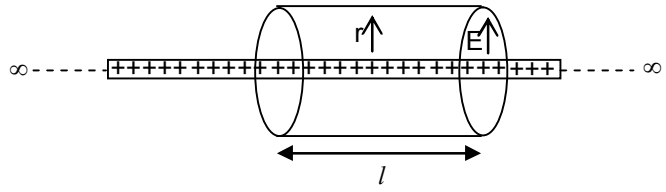
15. Relative velocity is $(v + v_0)$

Opponent frequency is $\frac{(v + v_0)}{\lambda}$

The no. of positive crests striking per sec. is same as frequency.

In 3 sec. $3\left(\frac{v + v_0}{\lambda}\right)$

16. From symmetry, $\vec{E}$ due to a uniform linear charged can only be radially directed. As a Gaussian surface, we can choose a circular cylinder of radius r and length. L, closed at each end by plane caps normal to the axis.



$\varepsilon_o \oint \vec{E}.\, d\vec{s} = q_{in}$

$\varepsilon_o \left[\int \vec{E}.d\vec{s} + \int \vec{E}.d\vec{s}\right] = q_{in}$

Cylindrical Plane Surface

$\varepsilon_o E(2\pi rl) + E.ds\ \cos 90^o = \lambda l$

$E = \dfrac{\lambda l}{\varepsilon_o 2\pi rl} = \dfrac{\lambda}{2\pi\varepsilon_0 r}$

The direction of $\vec{E}$ is radially outwards for a line of positive charge.

17. $E_x = -\dfrac{\partial V}{\partial x} = -(10x + 5y) = -10 + 10 = 0$

$E_y = -\dfrac{\partial V}{\partial y} = -5x = -5$

$\therefore \vec{E} = -5\hat{j}$ V/m.

18. Work done = change in potential energy

$= U_2 - U_1$

$U_1 = (1/2)E^2C$

$U_2 = \dfrac{1}{2}\dfrac{(EC)^2}{C'} = \dfrac{1}{2}\dfrac{E^2C^2}{C}\varepsilon_r = (1/2)E^2C\varepsilon_r$

$\therefore$ Work done = $(1/2)E^2C(\varepsilon_r - 1)$.

19. The resistance of the parallel combination of 2Ω and 3Ω resistors is given by $\dfrac{1}{R} = \dfrac{1}{2} + \dfrac{1}{3} = \dfrac{5}{6} \Rightarrow R = 1.2\Omega$

This resistance is in series with 2.8Ω giving a total effective resistance $= 1.2 + 2.8\Omega = 4\Omega$.

In the steady state, charge on the capacitor C has stablised and hence no current passes through 4Ω resistor which is in series with the capacitor.

Thus the current through the circuit = 6/4 = 1.5 A,

$V_{AB} = 1.5 \times 1.2 = 1.8$V, I through 2Ω resistor = 1.8/2 = 0.9 A.

20. $r = 1.5\Omega,\ l_1 = 52$cm, $l_2 = 40$cm

$\dfrac{r}{R} = \dfrac{l_1 - l_2}{l_2}$

resistance R = $\dfrac{rl_2}{(l_1 - l_2)}$

$$= \frac{1.5 \times 40}{(52-40)}$$

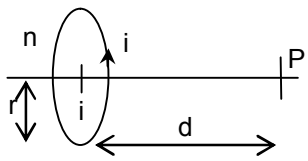
$$= \frac{1.5 \times 40}{12} = 5\Omega$$

21. $R = \frac{\rho l}{A} = \frac{\rho l^2}{Al} = \frac{\rho l^2}{V} = \frac{\rho l^2}{m/d} = \frac{\rho d l^2}{m}$

or $R \propto \frac{l^2}{m}$

$$R_1 : R_2 : R_3 = \frac{l_1^2}{m_1} : \frac{l_2^2}{m_2} : \frac{l_3^2}{m_3} = \frac{25}{1} : \frac{9}{3} : \frac{1}{5} = 125 : 15 : 1$$

22. The magnetic field at P due to the flat coil of n turns, radius r, carrying current i is



$$B = \frac{\mu_0}{2} \cdot \frac{nir^2}{\left(d^2 + r^2\right)^{3/2}} \cong \frac{\mu_0}{2} \cdot \frac{nir^2}{d^3} \left(d >> r\right)$$

$$= \frac{\mu_0}{2\pi} \cdot \frac{n\left(\pi r^2\right)i}{d^3} = \frac{\mu_0}{2\pi} \cdot \frac{\mu}{d^3}.$$

23. $B = \frac{\mu_0 Ni}{l}$ where N = Total number of turns, l = length of the solenoid

$$\Rightarrow 0.2 = \frac{4\pi \times 10^{-7} \times N \times 10}{0.8}$$

$$\Rightarrow N = \frac{4 \times 10^4}{\pi}$$

Since N turns are made from the winding wire so length of the wire

$(L) = 2\pi r \times N[2\pi r =$ length of each turns$]$

$$\Rightarrow L = 2\pi \times 3 \times 10^{-2} \times \frac{4 \times 10^4}{\pi} = 2.4 \times 10^3 m$$

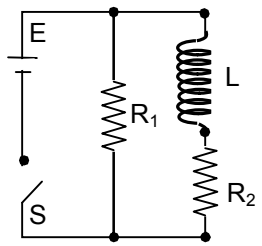
24. $I_1 = \frac{E}{R_1} = \frac{12}{2} = 6A$

$$E = L\frac{dI_2}{dt} + R_2 \times I_2$$

$$I_2 - I_0\left(1 - e^{-t/t_0}\right) \Rightarrow I_0 = \frac{E}{R_2} = \frac{12}{2} = 6A$$

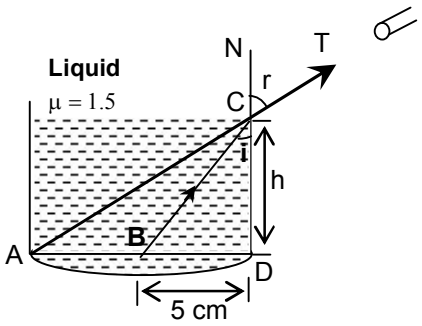
$$t_0 = \frac{L}{R} = \frac{400 \times 10^{-3}}{2} 0.2$$

$$I_2 = 6\left(1 - e^{-t/0.2}\right)$$



Potential drop across L = E - $R_2 I_2 = 12 - 2 \times 6\left(1 - e^{-bt}\right) = 12\ e^{-5t}$

25. It is mentioned in the problem that on filling the vessel with the liquid, point B is observed for the same setting; this means that the image of point B, is observed at A, because of refraction of the ray at C. For refraction at C,



$$\frac{\sin r}{\sin i} = {}_a\mu_l = 1.5$$

Now,

$$\sin r = \frac{AD}{AC} = \frac{10}{\sqrt{10^2 + h^2}}$$

Where h is the height of vessel.

$$\sin i = \frac{BD}{BC} = \frac{5}{\sqrt{5^2 + h^2}}$$

$$\therefore \frac{10}{\sqrt{100 + h^2}} . \frac{\sqrt{25 + h^2}}{5} = 1.5$$

$$\therefore \frac{25 + h^2}{100 + h^2} = \frac{9}{16}$$

$$\therefore 400 + 16h^2 = 900 + 9\,h^2$$

$$\therefore 7h^2 = 500$$

$$\therefore h = 8.45 \text{ cm}$$

26. In this case

$$\frac{1}{f_L} = (\mu - 1)\left(\frac{1}{\infty} - \frac{1}{-R}\right) = \frac{(\mu - 1)}{R}$$

and $F_M = \dfrac{(-R)}{2}$

so $P_L = \dfrac{1}{f_L} = \dfrac{(\mu - 1)}{R}$

and $P_M = -\dfrac{1}{f_M} = \dfrac{2}{R}$

and hence power of system

$P = P_L + P_M + P_L = 2P_L + P_M$

$$\Rightarrow P = \frac{2(\mu - 1)}{R} + \frac{2}{R} = \frac{2\mu}{R}$$

$$\therefore F = -\frac{1}{P} = -\frac{R}{2\mu}$$

i.e. the lens will be equivalent to a converging mirror of focal length $(R / 2\mu)$

27. For eyepiece,

$v_E = -25$ cm, $f_E = +5$ cm

$\Rightarrow u_E = -4.17$ cm $\simeq -4.2$ cm

$L = v_0 + |u_E| = 12.2$ cm here $v_0 = 8$ cm

$f_0 = +1$ cm $\Rightarrow u_0 = -1.1$ cm

28. Wavelength associated with a particle is given by

$\lambda = \dfrac{h}{\sqrt{2mE}}$

$\lambda = \dfrac{h}{\sqrt{2meV}} = \sqrt{\dfrac{150}{V}}A^{o}$ $\left(\text{for electron M} = 9.11\times10^{-31}\text{kg}\right)$

$\lambda = \sqrt{\dfrac{150}{10\times10^{3}}} = 0.122\ A^{o}$

For minima

$d\sin\theta = n\,\lambda$

For first minima n = 1

$0.55\sin\theta = 0.122$

$\sin\theta = \dfrac{0.122}{0.55} = 0.2218$

29. $hv = hv_0 + E$

$E = hv - hv_0$

But $v = \dfrac{c}{\lambda}$

$E = hc\left(\dfrac{1}{\lambda} - \dfrac{1}{\lambda_0}\right) = hc\left(\dfrac{\lambda_0 - \lambda}{\lambda_0\lambda}\right)$

30. The resistivity of pure Si is given by

$\rho = \dfrac{1}{\sigma} = \dfrac{1}{e\left(n_e\mu_e + n_h\mu_h\right)} = \dfrac{1}{en_1\left(\mu_e + \mu_h\right)}$

or $n_i = \dfrac{1}{e\rho\left(\mu_e + \mu_h\right)} = \dfrac{1}{1.6\times10^{-19}\times3000\left(0.12 + 0.045\right)}$

$= 1.26\times10^{16}\,m^{-3}$

When 10^{19} atoms of phosphorous (donor atoms of valence five) are added per m^3, the semiconductor becomes n – type semiconductor.

$\therefore n_e - n_h \approx n_e = N_d = 10^{19} \quad \because n_h = 1.26\times10^{16}$

Resitivity $\rho = \dfrac{1}{n_e\mu_e e} = \dfrac{1}{1.6\times10^{-19}\times10^{19}\times0.12} = 5.21\,\Omega m$

Chemistry

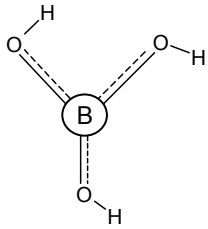
PART – II

SECTION – A

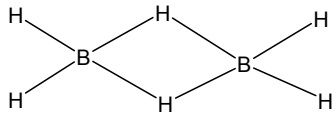
1. Aldehyde form 2° alcohol with RMgX.

2. It does not have α-hydrogen atom connected to sp^3 carbon.

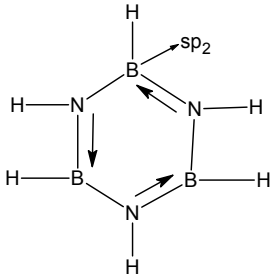
3. $H_3BO_3 \rightarrow$



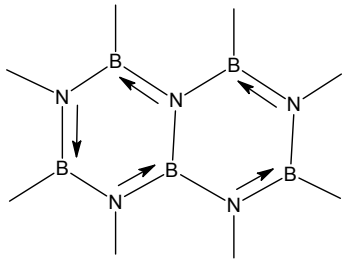
$B_2H_6 \rightarrow$



Borazine →



BN→



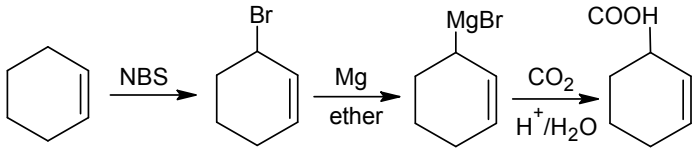
4. The cell reaction is

$H_2(g) + I_2(g) \rightleftharpoons 2H^+(aq) + 2I^-_{(aq)}$

$$0.7714 = 0.535 - \frac{0.0591}{2}\log\frac{\left[H^+\right]^2\left[I^-\right]^2}{p_{H_2}}$$

pH = 3

5.



6. $As_2S_3 \rightarrow H_3AsO_4 + H_2SO_4$

$(As^{3+})_2 \rightarrow 2As^{5+} + 4e^-$

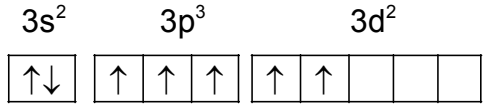
$(S^{2-})_3 \rightarrow 3S^{6+} + 24e^-$

$As_2S_3 \rightarrow 2As^{5+} + 3S^{6+} + 28e^-$

$E = M/28$

$Mn^{7+} + 5e^- \rightarrow Mn^{2+}$ $\quad \therefore E = M/5$

7. $Cl_{17} \rightarrow 3s^2\ 3p^5$
Cl_{17} (1st excited state) $3s^2\ 3p^4\ 3d^1$
Cl_{17} (2nd excited state)



n –	3	3	3	3	3
ℓ –	1	1	1	2	2
m_ℓ –	–1	+1	0	+2	+1

$\Sigma (n + \ell + m) = 25$

8. The overall rate constant, $K = K_1 + K_2 + K_3 = 6.93 \times 10^{-3}$

$$t_{1/2} = \frac{0.693}{6.93 \times 10^{-3}} = 100\,\text{sec}$$

After half life, total pressure, P = 4 atm

Now, $\frac{P_B}{P} = \frac{K_1}{K}$

$$P_B = \frac{2 \times 10^{-3} \times 4}{6.93 \times 10^{-3}} \Rightarrow 1.154\ \text{atm}$$

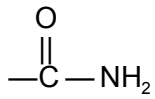
11. In osmois water flows from hypotonic solution towards hypertonic solution through SPM

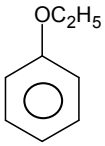
12. In glucose there are 5 – OH groups

13. In CuF_2, Copper is in +2 oxidation state

14. Hydro fluoric acid is not preserved in glass bottles because it reacts with silicon dioxide of glass

15. Galena is sulphide ore

17. $LiAlH_4$ does not reduce 

18. Phenetole is 

21. $4A + 2B + 3C \rightarrow A_4B_2C_3$
Given 1 mole 0.6 mol 0.72 mol ?
In the above, C is the limiting reagent hence it will give least product on complete consumption.
$\because$ 3 mol C give 1 mol product
$\therefore$ 0.72 mol C give 0.24 mol product

22. (A) Oxidation state of $T\ell$ in $T\ell I_3$ is +1

(B) $B_2O_3 + P_2O_5 \longrightarrow 2BPO_4$

(C) $B(OH)_3 + H_2O \rightleftharpoons \left[B(OH)_4\right]^- + H^+$

(D) $Na_2B_4O_7 + 7H_2O \longrightarrow \underset{\text{Salt}}{2Na\left[B(OH)_4\right]} + \underset{\text{Weak acid}}{2HBO_3}$

23. Na_2O has antiflourite structure in which O^{2-} ions are present at ccp and Na^+ ions are occupy all the tetrahedral voids which are located at body diagonals.

24. In both oxidation state of carbon is +2.

25. Vol. Strength of $H_2O_2 = 5.6 \times N$

$\therefore$ Normality of 10 vol = $\frac{10}{5.6}N$

Normality of 20 vol = $\frac{20}{5.6}N$

Normality of 15 vol $= \frac{15}{5.6}N$

$\therefore$ Total equivalent of $H_2O_2 = \frac{0.56\times10}{5.6} + \frac{0.56\times15}{5.6} + \frac{0.56\times20}{5.6}$ = 1 + 1.5 + 2.0 = 4.5

$\therefore N_{H_2O_2} = \frac{4.5}{3\times0.56} = 2.68$

26. By using the formula

$$2.303 \log \frac{P_2}{P_1} = \frac{\Delta H}{R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

For water boiling point = 373 K and thus P = 760 mm

Also $\Delta H = 540 \times 18$ cal/mol

$$2.303 \log \frac{760}{P_1} = \frac{540\times18}{2}\times\left[\frac{373-300}{373\times300}\right]$$

P_1 = 31.92 mm

27. O_2, O_2^+ and O_2^- all are paramagnetic in nature.

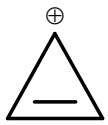
28. $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

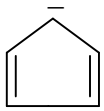
$CaO + H_2O \longrightarrow Ca(OH)_2$

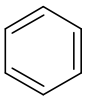
$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3$

$2CaCO_3 + 2H_2O + 2CO_2 \xrightarrow{\text{excess}} 2Ca(HCO_3)_2$

29. Option (b), (c), (e) & (f) shows syn addition.

30.  → sp^2 carbon, aromatic and diamagnetic

 → sp^2 carbon, aromatic and diamagnetic

 → sp^2 carbon, aromatic and diamagnetic

Mathematics

PART – III

SECTION – A

1. $P(\text{A wins}) = \frac{9}{14}$; $P(\text{B wins}) = \frac{5}{14}$

$P(\text{required}) = \frac{9}{14} \times \frac{2}{6} + \frac{5}{14} \times \frac{2}{4} = \frac{11}{28}$

2. General term in the expansion of $\left(\sqrt{3} + \sqrt[3]{2} + \sqrt[4]{5}\right)^8$ is $\frac{8!}{x!\, y!\, z!}\left(\sqrt{3}\right)^x \left(\sqrt[3]{2}\right)^y \left(\sqrt[4]{5}\right)^z$

(Where x + y + z = 8)
For rational term(s), we have the following cases:
(i) x = 2, y = 6, z = 0
(ii) x = 4, y = 0, z = 4
(iii) x = 0, y = 0, z = 8
(iv) x = 8, y = 0, z = 0

3. Let, $t = e^{|\sin x|} \in [1, e]$

$\Rightarrow t^2 + 4at + 1 = 0$
This should have two distinct roots in [1, e]

4. $\left(2^{\frac{\pi}{\sin^{-1}x}} - 4\right)\left(2^{\frac{\pi}{\sin^{-1}x}} - 2a\right) < 0$

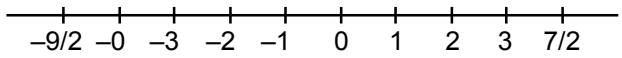
$2^{\frac{\pi}{\sin^{-1}x}} \in \left(0, \frac{1}{4}\right] \cup [4, \infty)$

5. Here, $f^2(x)\cot^2 x - 2f(x)\cot x + f^2(x) - \cos^2\theta = 0$ for which $D \geq 0$

6. $y = 7\cos x + 5\sin x = 8\left(\frac{7}{8}\cos x + \frac{5}{8}\sin x\right) = 8\sin(\alpha + x)$

$\Rightarrow \sin(\alpha + x) = \frac{2k+1}{\pm 8}$

$-1 \leq \frac{2k+1}{8} \leq 1$. $-1 \leq \frac{-2k+1}{8} \leq 1$, $-\frac{9}{2} \leq k \leq \frac{7}{2}$



Hence number of integral values are 8

7. Here, $x\sqrt{1-y^2} + y\sqrt{1-x^2} = z$

8. Variance = $\frac{\sum_{r=0}^{n} r^2 \cdot {}^nC_r}{\sum_{r=0}^{n} {}^nC_r} - \left(\frac{\sum_{r=0}^{n} r \cdot {}^nC_r}{\sum_{r=0}^{n} {}^nC_r}\right)^2$

10. For f(x) to be strictly decreasing $f'(x) = 3x^2 + 2ax + 2b < 0$ for some x
i.e., f(x) must have distinct real roots
Thus, $D > 0 \Rightarrow a^2 > 6b$
This includes 12 cases for
(a, b), i.e., (3, 1), (4, 1), (4, 2), (5, 1), (5, 2), (5, 3), (5, 4), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5)
Thus, $P(\text{required}) = \frac{1}{3}$

11. Let P(h, k) be the point of intersection
Then, equation of chord of contact is $ky - 2ax = 2a(h + 4a)$
Homogenising it with ellipse, $\frac{x^2}{4a^2} + \frac{y^2}{a^2} - \left\{\frac{ky - 2ax}{2a(h + 4a)}\right\}^2 = 0$
represents pair of straight lines from origin to AB which are perpendicular to each other

12. Here, $f(x) + f(2 - x) = 1$

13. Here, L.H.L. $= (-1)^{q-1}$ and R.H.L. $= (-1)^q$

14. Since the given function is not defined for any real number so its domain is ϕ so that the integration will not exist

16. Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{p}$ be position vectors of A, B, C and P respectively
Then, P.V. of orthocentre H = $\vec{a} + \vec{b} + \vec{c}$
P.V. of $M = \frac{\vec{a} + \vec{b} + \vec{c} + \vec{p}}{2}$
$\overrightarrow{DM} = \frac{\vec{a} + \vec{p}}{2}$
$\Rightarrow \overrightarrow{DM} \cdot \overrightarrow{PA} = 0$

17. Here, $f'(x) = x^{n-1}(x - 3)^2 [x(n + 3) - 3n]$
$\Rightarrow f'(x) = 0$ at $x = 0, 3, \frac{3n}{n + 3}$

18. Here, t is not defined at x = 2 and y is not defined at t = –2, 3
Now, t = –2 at $x = \frac{3}{2}$ and t = 3 at $x = \frac{7}{3}$

19. $\int \left(1 + x - x^{-1}\right) e^{x+x^{-1}} dx = \int e^{x+x^{-1}} dx + \int x\left(1 - \frac{1}{x^2}\right) e^{x+x^{-1}} dx$
$= \int e^{x+x^{-1}} dx + x \cdot e^{x+x^{-1}} - \int e^{x+x^{-1}} dx = x \cdot e^{x+x^{-1}} + c$

20. Let points be A(–2, 4), B(–1, 2), C(1, 2) and D(2, 4)
Let BE be required line such that DE = λ
Then area of quadrilateral BCDE = $\frac{1}{2} \times$ Area of trapezium ABCD

21. Let $\log_e 10 = x$. Then, $\sum_{r=0}^{n}(-1)^r\ {}^nC_r \frac{1+rx}{(1+nx)^r}$

$= \sum_{r=0}^{n}(-1)^r \cdot {}^nC_r\left(\frac{1}{1+nx}\right)^r - \frac{nx}{1+nx}\sum_{r=1}^{n}(-1)^{r-1} \cdot {}^{n-1}C_{r-1}\left(\frac{1}{1+nx}\right)^{r-1}$

$= \left(1-\frac{1}{1+nx}\right)^n - \frac{nx}{1+nx}\left(1-\frac{1}{1+nx}\right)^{n-1} = 0$

22. Here, $\frac{(b+c-a)+(c+a-b)}{2} > \{(b+c-a)(c+a-b)\}^{1/2}$

$\frac{(c+a-b)+(a+b-c)}{2} > \{(c+a-b)(a+b-c)\}^{1/2}$

and $\frac{(a+b-c)+(b+c-a)}{2} > \{(a+b-c)(b+c-a)\}^{1/2}$

Multiplying, $abc > (a+b-c)(b+c-a)(c+a-b)$

23. Let P and Q be $\left(at_1^2, 2at_1\right)$ and $\left(at_2^2, 2at_2\right)$ respectively for the parabola $y^2 = 4ax$

Then, $t_1t_2 = -1$

Also, R and T are $\left(-a, -\frac{2a}{t_1}\right)$ and $\left(-a, -\frac{2a}{t_2}\right)$ respectively

$\Rightarrow$ Slope of $RS = \frac{1}{t_1}$ and slope of $TS = \frac{1}{t_2}$

24. Here, $|z-3| \le 4$ is circle with centre at (3, 0) and radius 4
$|z-1| + |z+1|$ is ellipse with foci at (–1, 0) and (1, 0) and length of major axis 3

25. Let the four points be $\left(ct_1, \frac{c}{t_1}\right), \left(ct_2, \frac{c}{t_2}\right), \left(ct_3, \frac{c}{t_3}\right)$ and $\left(ct_4, \frac{c}{t_4}\right)$

$\Rightarrow \tan\alpha = \frac{1}{t_1^2};\ \tan\beta = \frac{1}{t_2^2};\ \tan\gamma = \frac{1}{t_3^2};\ \tan\delta = \frac{1}{t_4^2}$

Also, $\frac{1}{t_1t_2t_3t_4} = -1$

26. Equation of tangent to curve at (x, y) is $Y - y = \frac{dy}{dx}(X-x)$

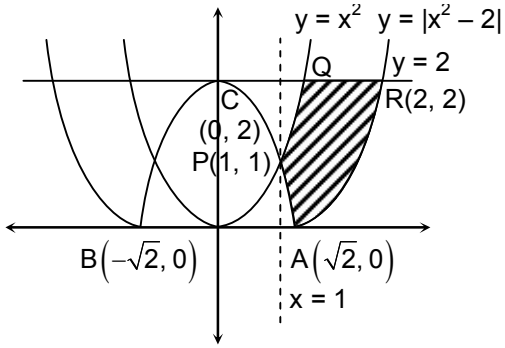
Let, lengths of perpendiculars be D_1 and D_2
Then $D_1^2 + D_2^2 = 2$

$\Rightarrow \left(\frac{dy}{dx}\right)^2\left(x^2-1\right) - 2xy\left(\frac{dy}{dx}\right) + y^2 = 0$

$\Rightarrow \frac{dy}{dx} = \frac{xy \pm y}{x^2-1}$

27. Required area
= Area of region APQA + Area of region AQRA

$$= \int_{1}^{\sqrt{2}} \left\{x^2 - \left(2 - x^2\right)\right\} dx + \int_{\sqrt{2}}^{2} \left(4 - x^2\right) dx$$



28. Here, $A + B + C = \pi$
$\Rightarrow e^{-i(B+C)} = -e^{iA}$; $e^{-i(C+A)} = -e^{iB}$; $e^{-i(A+B)} = -e^{iC}$

Now taking e^{iA}, e^{iB} and e^{iC} common from R_1, R_2 and R_3 $= -\begin{vmatrix} e^{iA} & -e^{iB} & -e^{iC} \\ -e^{iA} & e^{iB} & -e^{iC} \\ -e^{iA} & -e^{iB} & e^{iC} \end{vmatrix}$

29. Here, AQ = h

$BQ = \dfrac{h}{\sqrt{3}}$

Using cosine rule in ΔAQB

$h = 500\sqrt{\dfrac{3}{4-\sqrt{6}}}$ m

